Right Tube, Right Sample, Right Result:

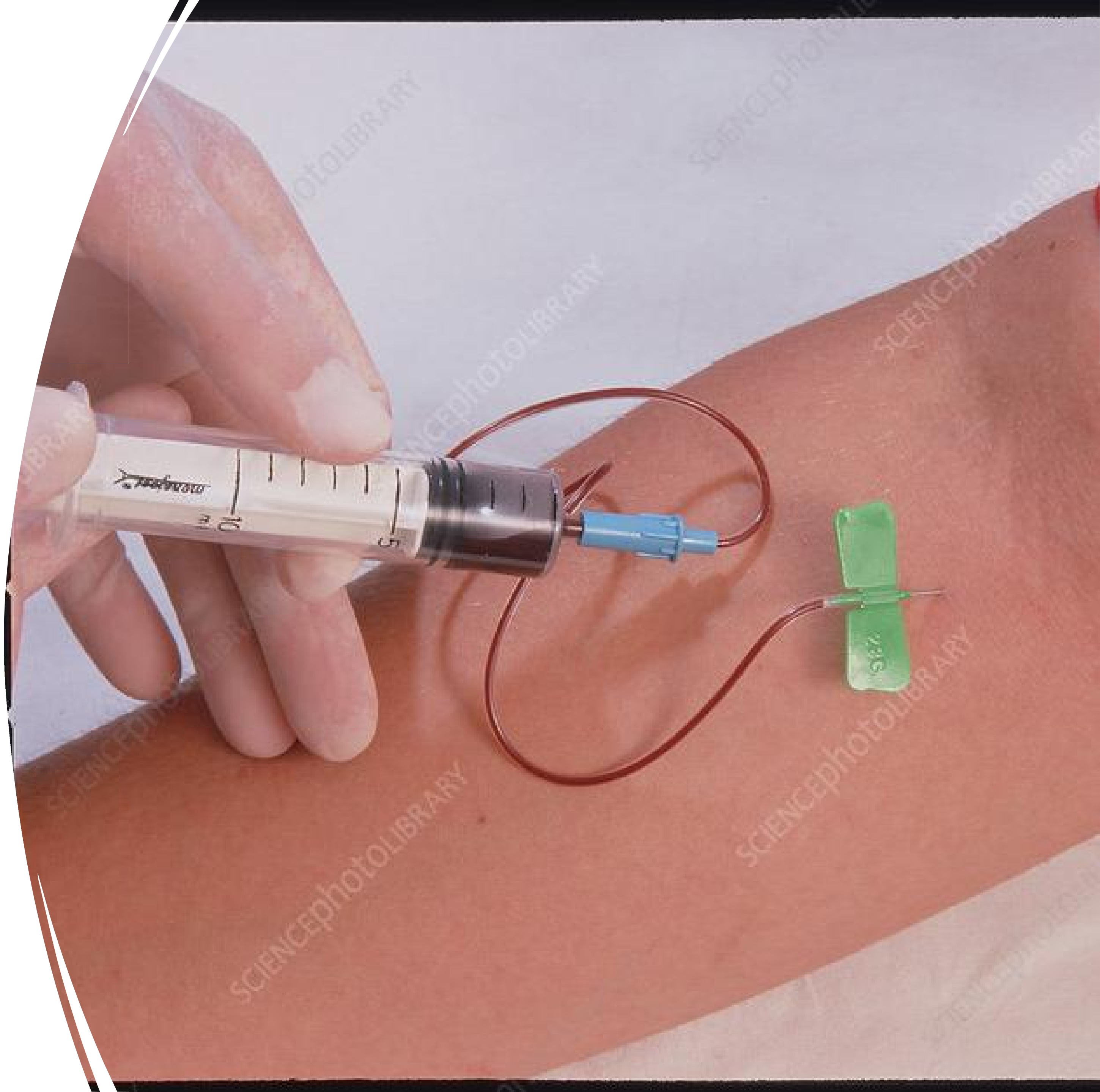
Practical Blood Sampling for Nurses

Why Blood Sampling by Nurses Matters

- Nurses are the **first** link between the patient and the laboratory result.
- Poor sample quality can cause up to 80–90% of laboratory errors in the pre-analytical phase (before the sample reaches the analyzer).
- One wrong tube, wrong label, or hemolyzed sample can lead to misdiagnosis, wrong dose, or delay in treatment.

Key message :

- “Every time you draw blood, you are directly affecting diagnosis and treatment, not just ‘filling tubes’.”



Common Nursing Errors in Blood Sampling

"Frequent Mistakes During Blood Collection"

- Wrong patient identification (*no name check, wrong label*).
- Wrong tube color for the requested test.
- Underfilled or overfilled tubes → *wrong blood: additive ratio, especially in light-blue citrate tubes.*
- Poor **blood draw** technique: *prolonged tourniquet time, vigorous "milking" or squeezing, using a too-small needle → hemolysis.*
- Not mixing tubes gently (or shaking hard) → *clots or hemolysis*.
- Taking blood from the same arm with running IV fluids → *diluted sample and false results*.

"Which of these errors do you think happens most in your hospital training?"

Short Scenario (Patient Harm)

"A Real-Lifestyle Scenario"

A 65-year-old diabetic patient in ER, suspected myocardial infarction.

Nurse takes blood from the arm with an IV dextrose infusion, into a gray tube instead of green, and squeezes the arm strongly.

Lab reports very high glucose and slightly abnormal potassium *because of hemolysis*.

Action: Doctor delays insulin adjustment, thinks glucose is extremely high due to disease, not the IV; potassium result is unreliable.

Result: delay in correct management and extra tests; patient stays longer, more cost, more risk.

Close the story :

"One small pre-analytical mistake created a big clinical problem."

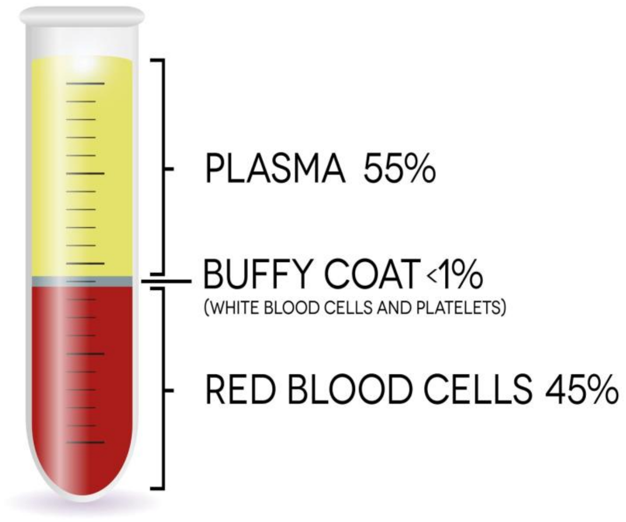
Brief on Blood Sampling & Pre-analytical Errors

"From Vein to Analyzer"

Correct patient ID → choose correct tube → proper venipuncture → correct fill volume → gentle mixing → timely transport to lab.

Highlight main pre-analytical problems that affect biochemistry:

- **Hemolysis (red-colored serum/plasma)**: *due to rough handling, small needle, shaking tubes, long tourniquet time → false K^+, LDH, AST, etc.*
- Delay in separation: *leaving whole blood for many hours → glucose decreases, some analytes change.*
- **Clotted anticoagulated samples**: *inadequate mixing of EDTA, citrate, or heparin tubes → unusable for many tests.*



Feature	Plasma	Serum
Anticoagulant used	Yes	No (blood clots)
Clotting factors	Present, fibrinogen	Mostly removed with clot
Separation time	Faster	Slower (needs clotting)

Blood Components, Plasma vs Serum

“What Is Inside the Tube?”

Whole blood = plasma + cells (RBCs, WBCs, platelets).

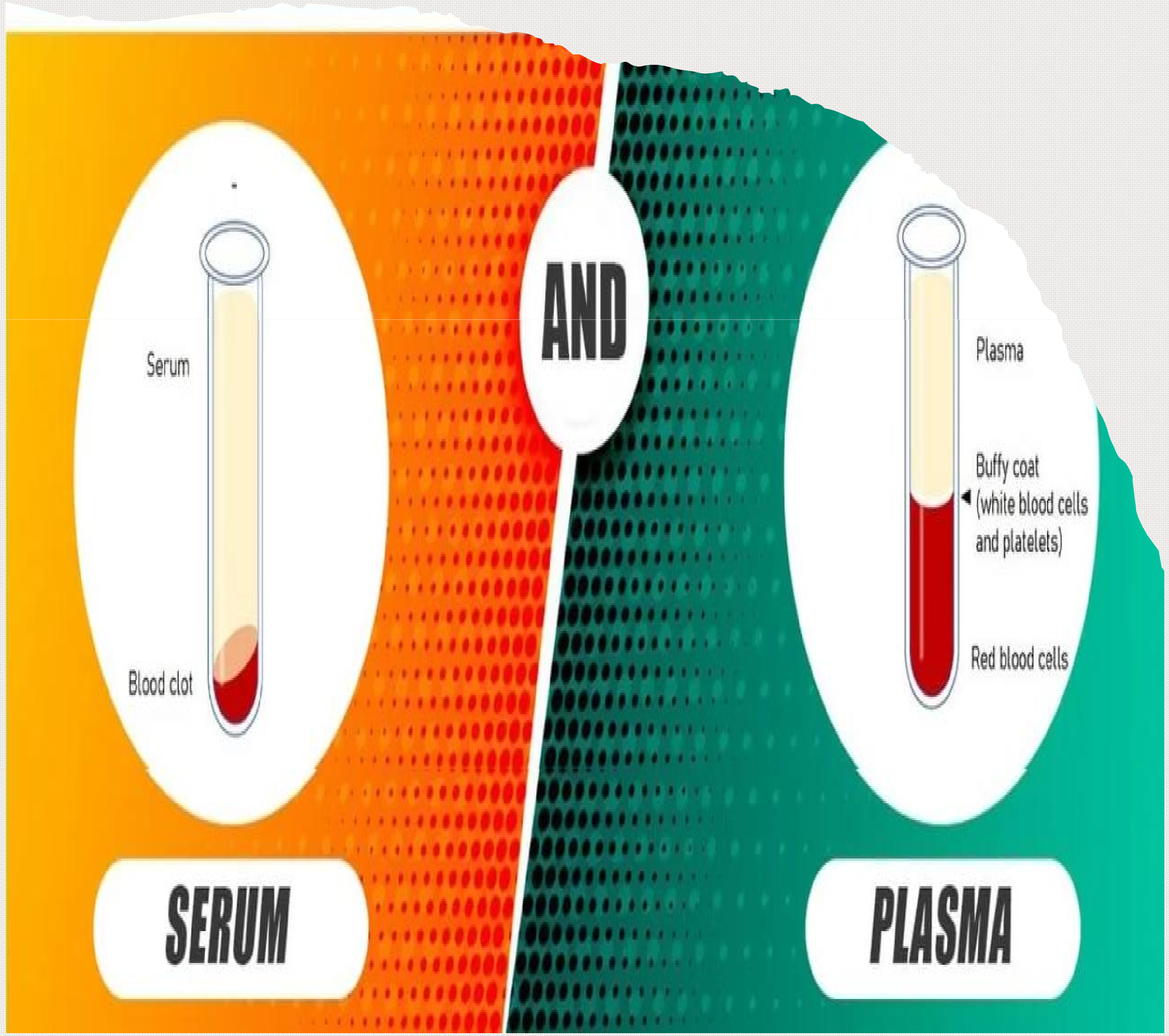
Plasma: liquid part of blood with anticoagulant present; still contains clotting factors, including fibrinogen.

Serum: liquid part after blood clots; no fibrinogen or clotting factors.

Think :

“If the tube has anticoagulant (like EDTA, citrate, heparin) and you centrifuge → *plasma*.”

“If the tube has no anticoagulant and blood is allowed to clot → *serum.*”



Common Tests on Plasma vs Serum

"Which Tests Use Plasma or Serum?"

Mostly serum (***red tube***): many biochemical, immunology, and serology tests:

- Liver enzymes (ALT, AST), kidney function (urea, creatinine), lipids, thyroid hormones, many antibodies.

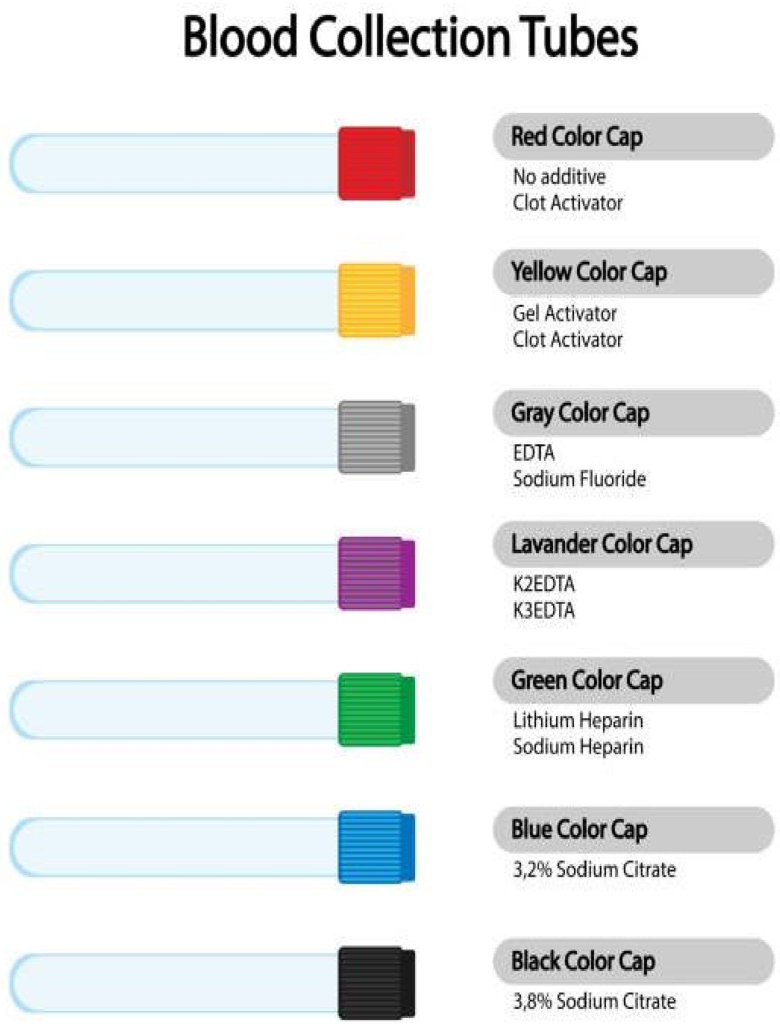
Mostly plasma (***e.g. light blue, green, some purple tubes***):

- Coagulation tests: PT, PTT, INR (citrate plasma).
- Some emergency biochemistry on heparin plasma (electrolytes, basic metabolic panel).

Can use either serum or plasma (results usually similar if pre-analytical steps correct):

- Many routine chemistry tests such as glucose, urea, creatinine, electrolytes, liver enzymes; labs choose based on protocol.

"For you as nurses: focus on using the tube requested, not choosing your own."



Tube Types and Six Common Colors

"Know Your Tubes and Colors"

Tube color = additive inside = type of sample (serum/plasma/whole blood) and type of tests.

(Color (top	Additive	Sample type	Typical use
Red	No additive or clot activator	Serum	Many biochemistry, serology, immunology tests.
(Gold / Yellow (SST	Gel + clot activator	(Serum (separated by gel	Routine chemistry, hormones
Light Blue	Sodium citrate	Citrated plasma	Coagulation: PT, PTT, INR (must be properly filled
Green	Lithium or sodium heparin	Heparinized plasma	Emergency chemistry, some routine tests.
Lavender / Purple	EDTA	Whole blood or plasma	CBC, blood film; some HbA1c.
Gray	Fluoride + oxalate	Fluoride plasma	Glucose (inhibits glycolysis

There are other colors (black for ESR, dark blue for trace elements, etc.), but these six are the most common they will meet now.

Fast Quiz

<u>Which tube color used for?”</u>

- “CBC → ....
- “PT and INR?” → ....
- “Fasting blood glucose?” → Gray (fluoride), but some labs accept serum/plasma if processed quickly..
- “Liver function tests?” → ....
- “Electrolytes in emergency room?” → ....
- What is the difference between plasma and serum?
- Mention the Common Nursing Errors in Blood Sampling
- What are the main pre-analytical problems that affect biochemistry?


Thank you